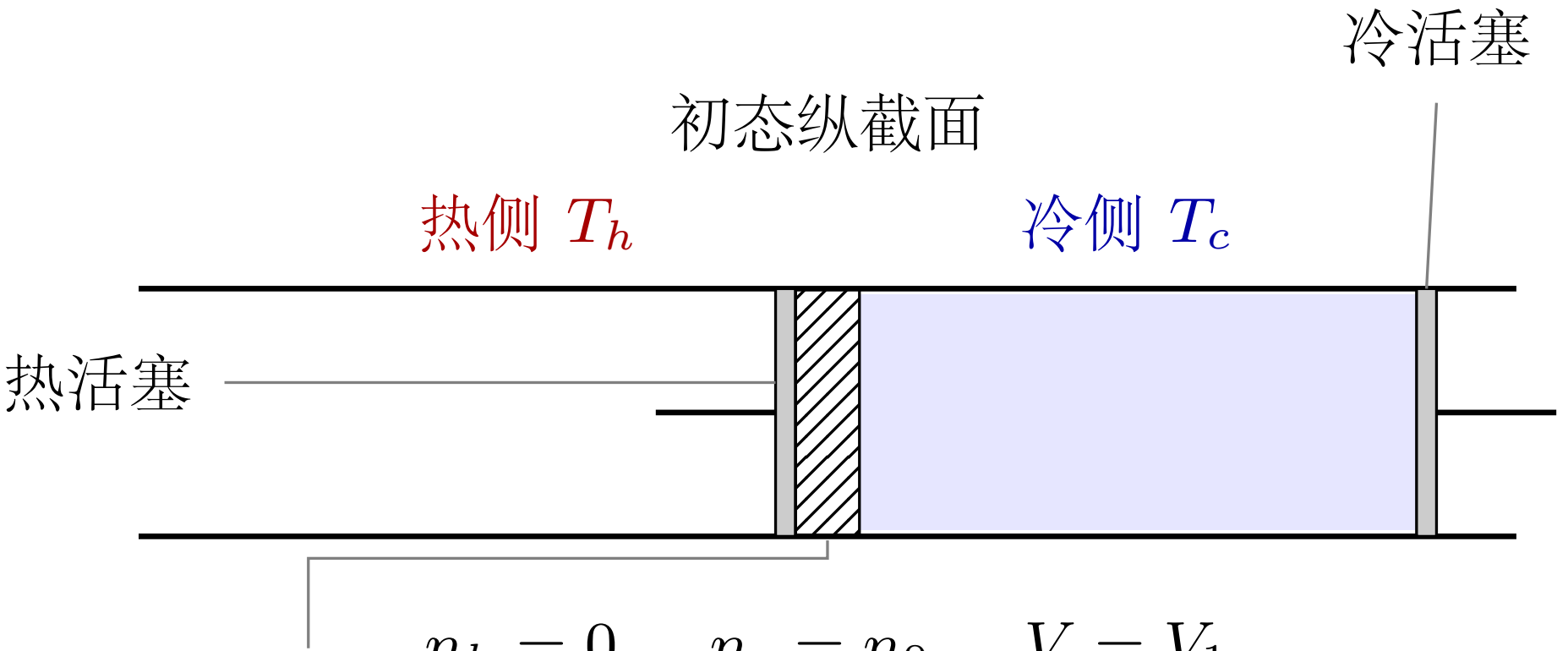
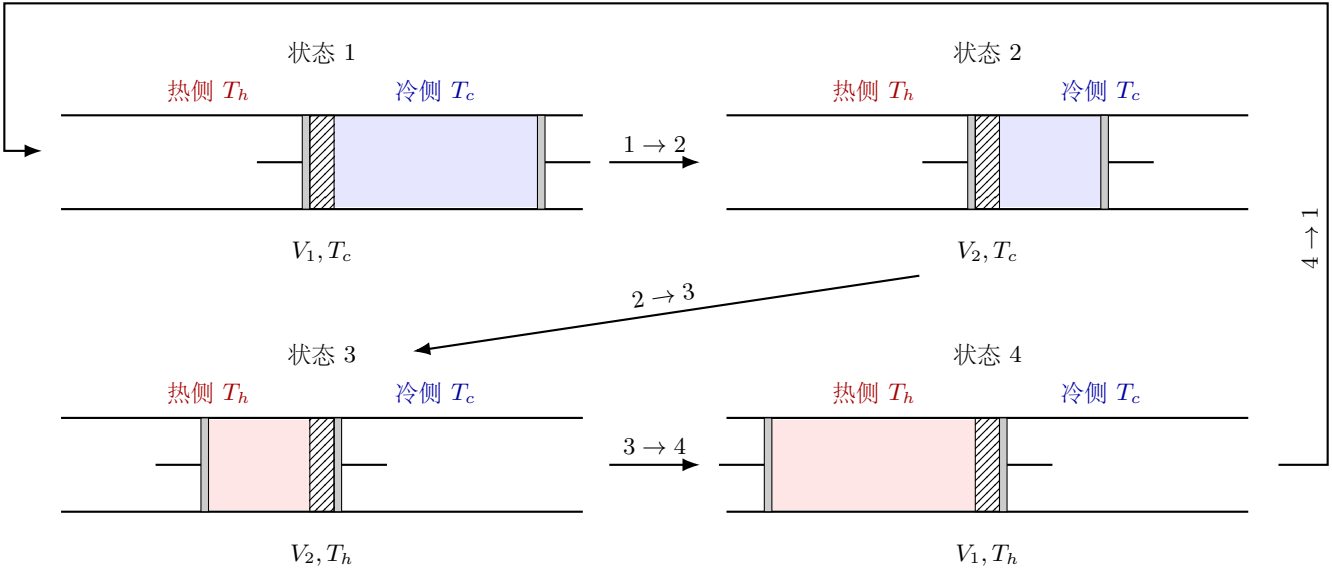
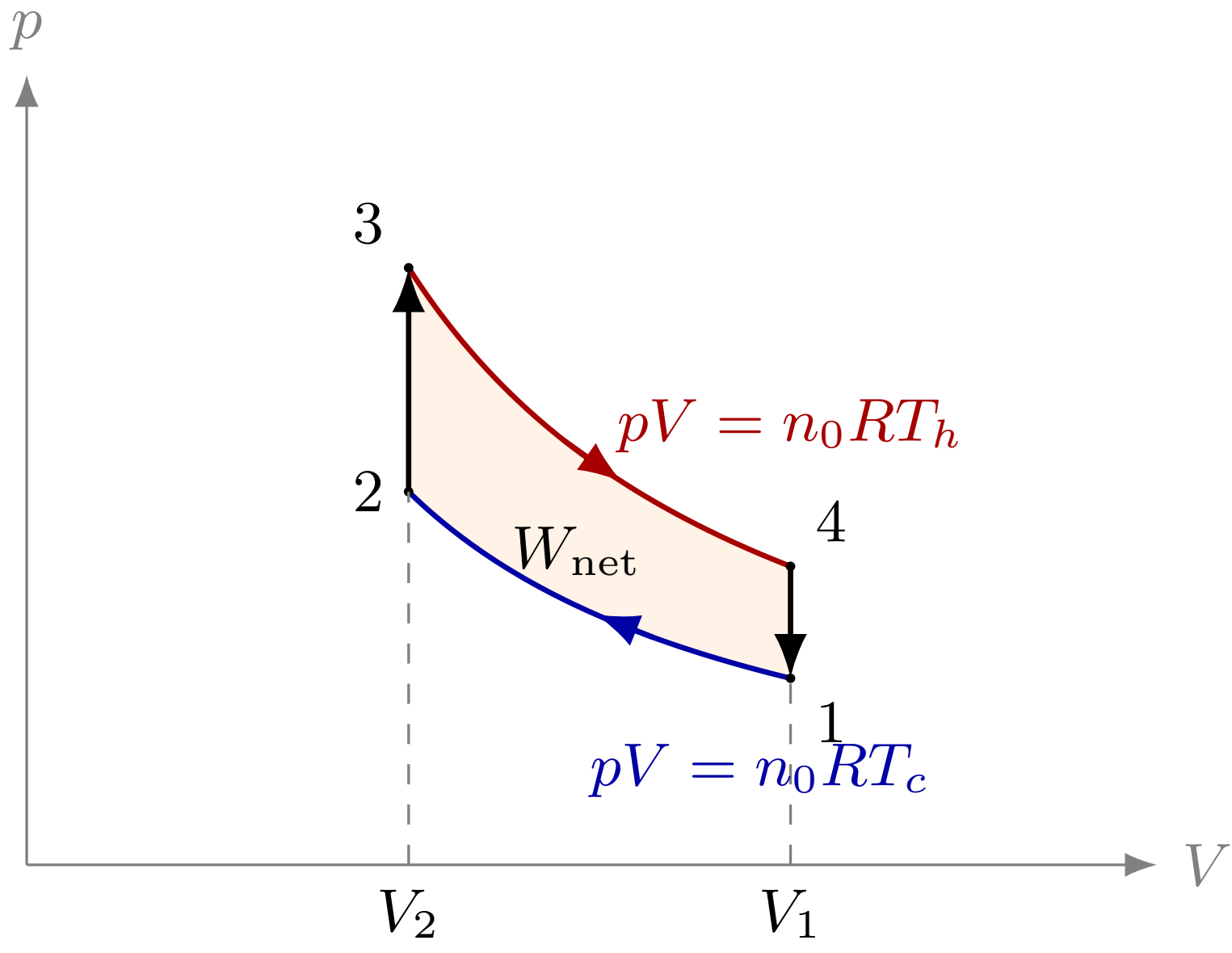
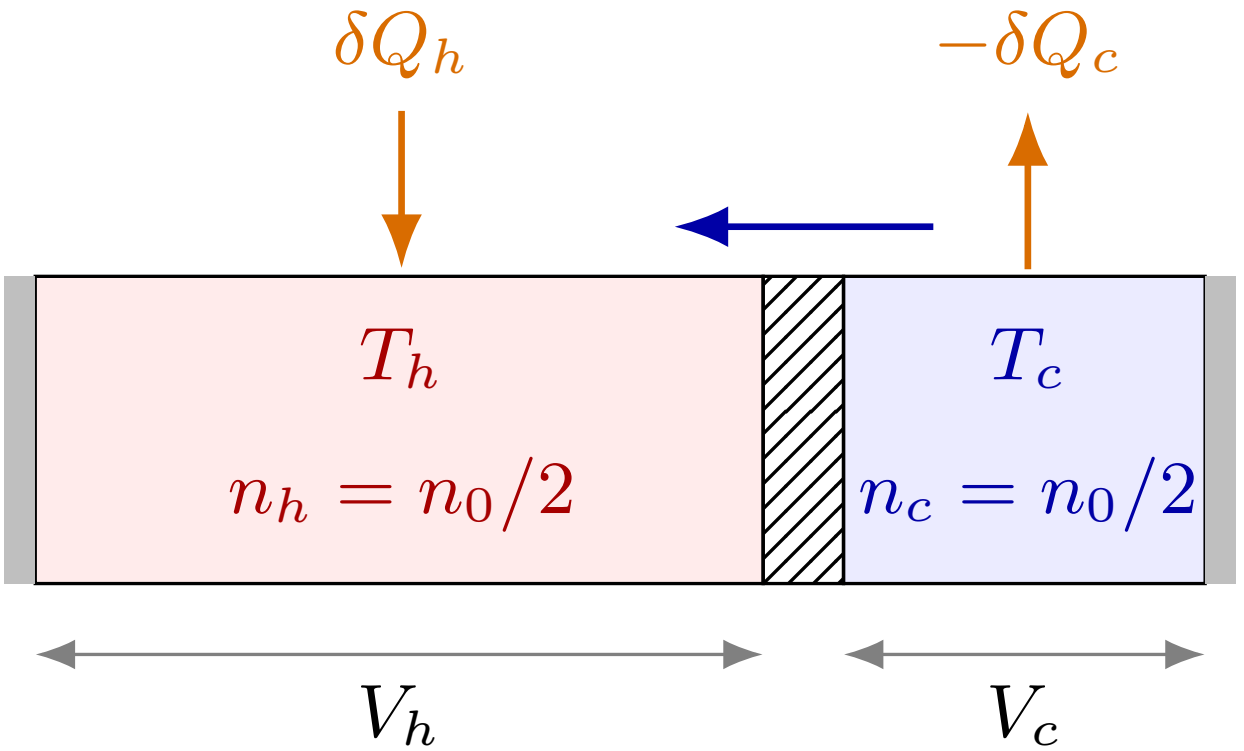
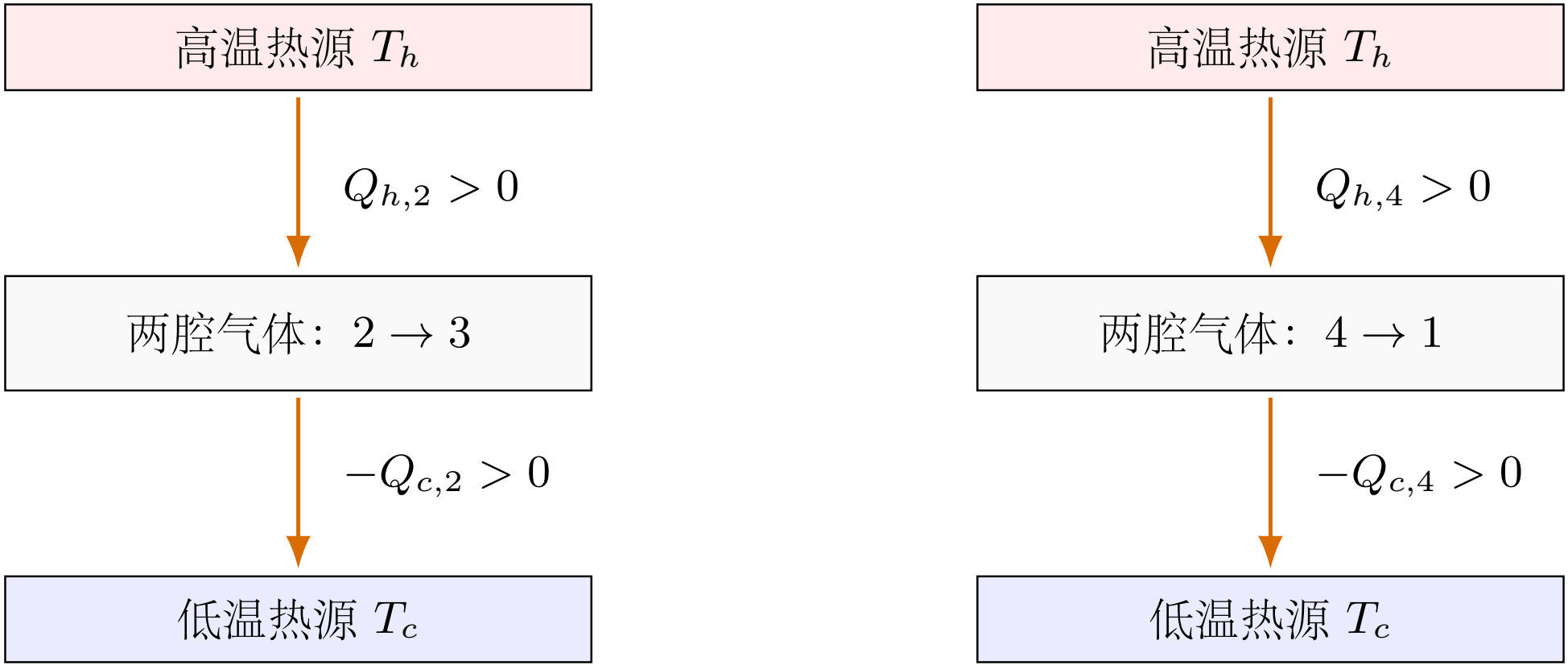

冷活塞
初态纵截面
热侧 T_h
冷侧 T_c
热活塞
$n_h = 0, \quad n_c = n_0, \quad V = V_1$
多孔通道
（关闭）


状态 1
热侧 T_h
冷侧 T_c
V_1, T_c
$1 \to 2$
状态 2
热侧 T_h
冷侧 T_c
V_2, T_c
$2 \to 3$
状态 3
热侧 T_h
冷侧 T_c
V_2, T_h
$3 \to 4$
状态 4
热侧 T_h
冷侧 T_c
V_1, T_h
$4 \to 1$


p
V
3
2
4
1
$pV = n_0 RT_h$
W_{net}
$pV = n_0 RT_c$
V_2
V_1



同一压力 p，两腔温度不同



无储热多孔塞模型；热源供热不等于两腔净吸热